INTERNAL ASSIGNMENT

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SET – 1

QUESTION -1: EVALUATE THE FOLLOWING LIMITS:

(I) $\lim_{n\to\infty}\left(\frac{n^2+4n+7}{2+3n+4n^2}\right)$ **(II)** $\lim_{x\to\pi/2}(tanx - secx)$.

ANSWER – **(i) lim n→∞ (n² + 4n + 7) / (2 + 3n + 4n²)**

This is a limit of the form ∞/∞, where both numerator and denominator are polynomials of the **same degree 2**. The standard method is to divide every term by the highest power of n occurring in the expression, which is **n²**.

Dividing numerator and denominator by n²:

= lim n→∞ [(n²/n² + 4n/n² + 7/n²) / (2/n² + 3n/n² + 4n²/n²)]

= lim n→∞ [(1 + 4/n + 7/n²) / (2/n² + 3/n + 4)]

Now apply the standard result **lim n→∞ (1/n^k) = 0 for k > 0**:

- 4/n → 0, 7/n² → 0
- 2/n² → 0, 3/n → 0

Therefore the limit becomes:

= (1 + 0 + 0) / (0 + 0 + 4) = **1/4**

Answer: lim n→∞ (n² + 4n + 7)/(2 + 3n + 4n²) = 1/4

Verification by the coefficient rule: when numerator and denominator have equal degree, the limit is the ratio of leading coefficients = 1/4. ✓

(ii) lim x→π/2 (tan x − sec x)

Write in sine and cosine form:

= lim x→π/2 [(sin x / cos x) − (1 / cos x)] = lim x→π/2 (sin x − 1)/cos x

As x → π/2, sin x → 1 and cos x → 0, so this is the indeterminate form **0/0**. Two methods:

Method 1 — Rationalisation (elegant, no calculus):

Multiply numerator and denominator by (sin x + 1):

= lim (sin x − 1)(sin x + 1) / [cos x (sin x + 1)] = lim (sin²x − 1) / [cos x (sin x + 1)]

Since sin²x − 1 = −cos²x:

= lim (−cos²x) / [cos x (sin x + 1)] = lim (−cos x)/(sin x + 1)

Putting $x = \pi/2$: $= (-0)/(1 + 1) = \mathbf{0}$
Method 2 — L'Hôpital's Rule: differentiating numerator and denominator,
$= \lim (\cos x)/(-\sin x) = 0/(-1) = \mathbf{0}$
Answer: $\lim_{x \to \pi/2} (\tan x - \sec x) = 0$

QUESTION -2: State Mean Value Theorem and verify it for $f(x) = x^2$ in [1, 3].
ANSWER –**Statement of Mean Value Theorem (Lagrange's Mean Value Theorem)**
If a function f(x) satisfies the following two conditions:

1. **f(x) is continuous** on the closed interval **[a, b]**, and
2. **f(x) is differentiable** on the open interval **(a, b)**,

then there exists **at least one point c in the open interval (a, b)** such that
$f'(c) = [f(b) - f(a)] / (b - a)$
Geometrical interpretation: the theorem states that for a smooth curve joining two points A(a, f(a)) and B(b, f(b)), there is at least one point on the curve where the **tangent is parallel to the chord AB**. The left-hand side f′(c) is the slope of the tangent at c, and the right-hand side is the slope of the chord AB.

Verification for $f(x) = x^2$ on [1, 3]
Here $f(x) = x^2$, a = 1, b = 3.
Step 1 — Check the conditions. $f(x) = x^2$ is a polynomial, and every polynomial is continuous and differentiable for all real x. Hence f(x) is **continuous on [1, 3]** and **differentiable on (1, 3)**. Both conditions of the theorem are satisfied.
Step 2 — Compute f(a) and f(b).

- $f(1) = 1^2 = 1$
- $f(3) = 3^2 = 9$

Step 3 — Compute the slope of the chord.
$[f(b) - f(a)]/(b - a) = (9 - 1)/(3 - 1) = 8/2 = \mathbf{4}$
Step 4 — Find f′(x) and equate.
$f'(x) = d/dx\ (x^2) = 2x$
We require $f'(c) = 4$:
$2c = 4 \Longrightarrow \mathbf{c = 2}$
Step 5 — Check that c lies in the interval. $c = 2$, and $2 \in (1, 3)$. ✓
Conclusion: the value c = 2 lies in the open interval (1, 3) and satisfies $f'(c) = 2(2) = 4 = [f(3) - f(1)]/(3 - 1)$. Hence the **Mean Value Theorem is verified** for $f(x) = x^2$ on [1, 3].

QUESTION -3: Find the value of tan 75° and sin 15°.
ANSWER – **(a) Value of tan 75°**
Write 75° as the sum of two standard angles: **75° = 45° + 30°**.
Using the compound-angle formula:
$\tan(A + B) = (\tan A + \tan B)/(1 - \tan A \cdot \tan B)$
Put A = 45°, B = 30°, with tan 45° = 1 and tan 30° = $1/\sqrt{3}$:

tan 75° = [1 + (1/√3)] / [1 − 1·(1/√3)]
Multiply numerator and denominator by √3 to simplify:
= (√3 + 1)/(√3 − 1)
Rationalise the denominator by multiplying by (√3 + 1)/(√3 + 1):
= (√3 + 1)(√3 + 1) / [(√3 − 1)(√3 + 1)]
= (√3 + 1)² / (3 − 1) [using (a−b)(a+b) = a² − b²]
= (3 + 2√3 + 1)/2 [using (a+b)² = a² + 2ab + b²]
= (4 + 2√3)/2
tan 75° = 2 + √3 ≈ 3.732 ✓

(b) Value of sin 15°
Write 15° as the difference of two standard angles: **15° = 45° − 30°**.
Using the compound-angle formula:
sin(A − B) = sin A·cos B − cos A·sin B
Put A = 45°, B = 30°, with sin 45° = cos 45° = 1/√2, sin 30° = 1/2, cos 30° = √3/2:
sin 15° = sin 45°·cos 30° − cos 45°·sin 30°
= (1/√2)(√3/2) − (1/√2)(1/2)
= √3/(2√2) − 1/(2√2)
= (√3 − 1)/(2√2)
Rationalise by multiplying numerator and denominator by √2:
= √2(√3 − 1)/(2 × 2) = **(√6 − √2)/4**
sin 15° = (√6 − √2)/4 ≈ 0.2588 ✓

SET – 2

QUESTION -4: Decompose $\frac{x^2}{(x-1)(x+2)(x-3)}$ into partial fractions.

ANSWER - **Step 1 — Check that the fraction is proper**
Numerator: x^2 → degree = 2
Denominator: (x−1)(x+2)(x−3) → degree = 3 (three linear factors)
Since **degree of numerator (2) < degree of denominator (3)**, the fraction is **proper**, so no long division is required and we may proceed directly to partial fractions.
Step 2 — Set up the partial fraction form
The denominator has **three distinct linear factors**: (x−1), (x+2), (x−3). Each distinct linear factor gets a **constant numerator**:
x²/[(x−1)(x+2)(x−3)] = **A/(x−1) + B/(x+2) + C/(x−3)** … (1)
where A, B, C are constants to be determined.
Step 3 — Clear the denominators
Multiply both sides of (1) by the LCM, i.e. (x−1)(x+2)(x−3):
x² = A(x+2)(x−3) + B(x−1)(x−3) + C(x−1)(x+2) … (2)
Equation (2) is an **identity** — it is true for every value of x. Hence we may substitute convenient values of x to isolate each constant.
Step 4 — Determine A by putting x = 1

Put x = 1 in (2). The second and third terms vanish (since they contain the factor (x−1)):
$1^2=A(1+2)(1-3)$
$1=A(3)(-2)$
$1=-6A$
A = −1/6
Step 5 — Determine B by putting x = −2
Put x = −2 in (2). The first and third terms vanish (they contain the factor (x+2)):
$(-2)^2=B(-2-1)(-2-3)$
$4=B(-3)(-5)$
$4=15B$
B = 4/15
Step 6 — Determine C by putting x = 3
Put x = 3 in (2). The first and second terms vanish (they contain the factor (x−3)):
$3^2=C(3-1)(3+2)$
$9=C(2)(5)$
$9=10C$
C = 9/10

Step 7 — Write the required decomposition
$x^2/[(x-1)(x+2)(x-3)] = -1/[6(x-1)] + 4/[15(x+2)] + 9/[10(x-3)]$

Step 8 — Verification (three independent checks)
Check 1 — Coefficient of x^2: expanding (2), the x^2 coefficient on the RHS is A + B + C, and on the LHS it is 1.
$A + B + C = -1/6 + 4/15 + 9/10 = (-5 + 8 + 27)/30 = 30/30 =$ **1** ✓

Check 2 — Coefficient of x: expanding each product,
- $A(x+2)(x-3) = A(x^2 - x - 6) \rightarrow$ x-coefficient = −A
- $B(x-1)(x-3) = B(x^2 - 4x + 3) \rightarrow$ x-coefficient = −4B
- $C(x-1)(x+2) = C(x^2 + x - 2) \rightarrow$ x-coefficient = +C

Sum = $-A - 4B + C = 1/6 - 16/15 + 9/10 = (5 - 32 + 27)/30 =$ **0** ✓ (LHS has no x term)

Check 3 — Constant term:
Constant term on RHS = $A(2)(-3) + B(-1)(-3) + C(-1)(2) = -6A + 3B - 2C = -6(-1/6) + 3(4/15) - 2(9/10) = 1 + 4/5 - 9/5 = 1 - 1 =$ **0** ✓ (LHS constant = 0)
All three checks hold, so the decomposition is correct.

Step 9 — Combined form
Taking LCM 30 in the answer:
$= [-5(x+2)(x-3) + 8(x-1)(x-3) + 27(x-1)(x+2)] / [30(x-1)(x+2)(x-3)]$
Numerator $= -5(x^2 - x - 6) + 8(x^2 - 4x + 3) + 27(x^2 + x - 2) = (-5x^2 + 5x + 30) + (8x^2 - 32x + 24) + (27x^2 + 27x - 54)$

$=30x^2+(5-32+27)x+(30+24-54)$
$= \mathbf{30x^2 + 0 + 0 = 30x^2}$

Hence the combined fraction = $30x^2/[30(x-1)(x+2)(x-3)] = \mathbf{x^2/[(x-1)(x+2)(x-3)]}$ ✓ — which is exactly the original expression. This confirms the result completely.

Final Answer:
$\mathbf{x^2/[(x-1)(x+2)(x-3)] = -1/[6(x-1)] + 4/[15(x+2)] + 9/[10(x-3)]}$

QUESTION -5: Find the equation of the tangent and normal line to the curve $y = x^3 - 3x^2 + x - 1$ at the point where $x = 3$.
ANSWER –**Step 1 — Find the point of contact.**
Substitute $x = 3$ in the equation of the curve:
$y = 3^3 - 3(3^2) + 3 - 1 = 27 - 27 + 3 - 1 = \mathbf{2}$
So the point of contact is **P(3, 2)**.
Step 2 — Find the slope of the tangent.
The slope of the tangent at any point is given by dy/dx. Differentiating the curve with respect to x:
$dy/dx = d/dx\,(x^3) - 3 \cdot d/dx\,(x^2) + d/dx\,(x) - d/dx\,(1)$
$dy/dx = \mathbf{3x^2 - 6x + 1}$
At $x = 3$:
$m = 3(3^2) - 6(3) + 1 = 27 - 18 + 1 = \mathbf{10}$
So the **slope of the tangent is m = 10**.
Step 3 — Equation of the tangent.
Using the point–slope form $\mathbf{y - y_1 = m(x - x_1)}$ with $(x_1, y_1) = (3, 2)$ and $m = 10$:
$y-2=10(x-3)$
$y-2=10x-30$
$\mathbf{y = 10x - 28}$
In general form: $\mathbf{10x - y - 28 = 0}$
Step 4 — Slope of the normal.
The normal is perpendicular to the tangent at the point of contact, so its slope is the **negative reciprocal** of the tangent's slope:
$m_{normal} = -1/m = \mathbf{-1/10}$
Step 5 — Equation of the normal.
Again using the point–slope form with (3, 2) and slope −1/10:
$y - 2 = (-1/10)(x - 3)$
Multiply throughout by 10:
$10y-20=-(x-3)$
$10y-20=-x+3$
$\mathbf{x + 10y - 23 = 0}$
Step 6 — Verification.

- Check the tangent passes through (3, 2): $10(3) - 2 - 28 = 30 - 30 = 0$ ✓
- Check the normal passes through (3, 2): $3 + 10(2) - 23 = 3 + 20 - 23 = 0$ ✓
- Check perpendicularity: $m_1 \times m_2 = 10 \times (-1/10) = -1$ ✓

Answer:

- **Equation of the tangent: y = 10x − 28, i.e. 10x − y − 28 = 0**
- **Equation of the normal: x + 10y − 23 = 0**

QUESTION-6: FIND THE DERIVATIVE OF Y =ax^{nx}.

ANSWER - **Given:** y = ax^{nx}, where **a** and **n** are constants.
Here the **base is x** and the **exponent is nx**, which also depends on x. This is a function of the form **u^v**, where both the base and the exponent are variables. Hence the ordinary power rule (d/dx $x^n = n \cdot x^{n-1}$) cannot be applied directly. We use **logarithmic differentiation**. Since the exponent involves x, we take **x > 0** so that x^(nx) is defined and differentiable.
Step 1 — Take the natural logarithm of both sides
y = a·x^(nx)
Taking ln on both sides:
ln y = ln(a·x^(nx))
Using the product rule of logarithms, ln(m·n) = ln m + ln n:
ln y = ln a + ln(x^(nx))
Using the power rule of logarithms, ln(x^k) = k·ln x:
ln y = ln a + nx·ln x … (1)
Step 2 — Differentiate both sides with respect to x
Here **a** is a constant, so **ln a is constant** and its derivative is **0**.
For the term **nx·ln x**, apply the **product rule**, d/dx(u·v) = u·(dv/dx) + v·(du/dx), taking u = nx and v = ln x:

- d/dx (nx) = n
- d/dx (ln x) = 1/x

Therefore,
d/dx (nx·ln x) = (nx)(1/x) + (ln x)(n) = n + n·ln x = **n(1 + ln x)**
Now differentiate (1). On the left side, y is a function of x, so by the **chain rule**:
(1/y)·(dy/dx) = 0 + n(1 + ln x)
Step 3 — Solve for dy/dx
Multiply both sides by y:
dy/dx = y·n(1 + ln x)
Step 4 — Substitute back y = a·x^(nx)
dy/dx = a·x^(nx)·n·(1 + ln x)
Step 5 — Final result
dy/dx = a·n·x^(nx)·(1 + ln x)
which may also be written as **a·n·x^(nx)·(ln x + 1)**.

Verification (Method 2 — Exponential Form)

Write the function using the identity x = e^(ln x):
y = a·x^(nx) = a·e^(nx·ln x)
Differentiate using the chain rule, d/dx(e^u) = e^u·(du/dx), where u = nx·ln x:
dy/dx = a·e^(nx·ln x)·d/dx(nx·ln x)
= a·x^(nx)·n(1 + ln x)
dy/dx = a·n·x^(nx)(1 + ln x) ✓
Both methods give the **same result**, confirming the answer.

Check with a Special Case
Put a = 1 and n = 1, so y = x^x. The formula gives:
dy/dx = 1·1·x^x·(1 + ln x) = **x^x(1 + ln x)**
which is the standard well-known result for the derivative of x^x. ✓
Numerical check: take a = 1, n = 2, so y = x^(2x). At x = 1, the formula gives dy/dx = 1·2·1²·(1 + 0) = **2**. Computing y at x = 1 and x = 1.001 gives a slope of approximately **2.003**, which agrees. ✓

Final Answer:
d/dx [a·x^(nx)] = a·n·x^(nx)·(1 + ln x)